Q.1 create a tuple my_tuple=(10,20,30,40,50)

In [14]:
```
my_tuple =(10,20,30,40,50)
print("First element :-",my_tuple[0])
print("Last element:-",my_tuple[-1])
print("Length of my tuple :-",len(my_tuple))
print(my_tuple[1:3])
```

```
First element :- 10
Last element:- 50
Length of my tuple :- 5
(20, 30)
```

Q.2 Create a tuple fruits=("appple","banana","mango","orange")

In [17]:
```
fruits=("appple","banana","mango","orange")
print("Second fruit is:-",fruits[1])
print("Last two fruits:-",fruits[2:3])
print("total number of fruits:-",len(fruits))
```

```
Second fruit is:- banana
Last two fruits:- ('mango',)
total number of fruits:- 4
```

Q.3 Create a set numbers={10,20,30,40,50}

In [ ]:
```
numbers={10,20,30,40,50}
print("Complete set:-",numbers)
print("Length of my set:-",len(numbers))
print("")
```

```
Complete set:- {50, 20, 40, 10, 30}
Length of my set:- 5
```

Q.4 Create two sets set1={1,2,3,4} set2={3,4,5,6}

In [31]:
```
set1={1,2,3,4}
set2={3,4,5,6}
print("set1 U set2 :-",set1.union(set2))
print("Intersection of my sets:-",set1.intersection(set2))
```

```
set1 U set2 :- {1, 2, 3, 4, 5, 6}
Intersection of my sets:- {3, 4}
```

Q.5 Create a dictionary: student={"name":"Kriti","age":20,"course":"Python"}

In [34]:
```
student={"name":"Kriti","age":20,"course":"Python"}
print(student)
print(student['name'])
print(student["age"])
```

```
{'name': 'Kriti', 'age': 20, 'course': 'Python'}
Kriti
20
```

In [ ]:

```
numbers=[12,45,7,23,56,89,3
```

Q.6 Create a list arr=[10,20,30,40,40,60]

In [35]:

```
arr=[10,20,30,40,40,60]

def reverse_list(arr):
    return arr[::-1]

print(reverse_list(arr))
```

```
[60, 40, 40, 30, 20, 10]
```

Q.7 Create a tuple data={5,10,15,20,25,30,35}

In [38]:

```
data={5,10,15,20,25,30,35}

count=0
for i in data:
    if i%5 ==0:
        count=count+1

print("Element divible by 5 :-",count)
print("Sum of all element:-",sum(data))
print("Average:-",sum(data)/len(data))
```

```
Element divible by 5 :- 7
Sum of all element:- 140
Average:- 20.0
```

Q.8 Create a dictionary: students ={ "Aman":78, "Riya":92, "Kirti":88, "Rahul":95 }

In [40]:

```
students ={
"Aman":78,
"Riya":92,
"Kirti":88,
"Rahul":95
}
print("Student with highest marks:-",max(students, key=students.get))
print("Student with the lowest marks:-",min(students, key=students.get))

print("Student who scored more than 85 :-",)
for i in students:
    if students[i]>85:
        print(i ,students[i])
```

```
Student with highest marks:- Rahul
Student with the lowest marks:- Aman
Student who scored more than 85 :-
Riya 92
Kirti 88
Rahul 95
```

Q.9 Write a function count_frequency(arr) that takes a list as input and prints the frequency of each element

In [41]:
```
arr=[1,2,2,3,3,3,3,5,5,6,6,6]

def count_frequency(arr):
    for i in set(arr):
        print(i,"->",arr.count(i),"times")

count_frequency(arr)
```

```
1 -> 1 times
2 -> 2 times
3 -> 4 times
5 -> 2 times
6 -> 3 times
```

Q.10 Write a function: find_duplicates(arr)

that takes a list as input and prints all duplicate element present in the list

In [42]:
```
arr=[10,20,30,20,40,10,50,30]

def find_duplicates(arr):
    print("Duplicate elements are:")
    for i in set(arr):
        if arr.count(i)>1:
            print(i)

find_duplicates(arr)
```

```
Duplicate elements are:
10
20
30
```